

LOW HEIGHT WEIGHING ASSEMBLY FOR SILO, TANK AND VESSEL WEIGHING

capacities 1t - 30t

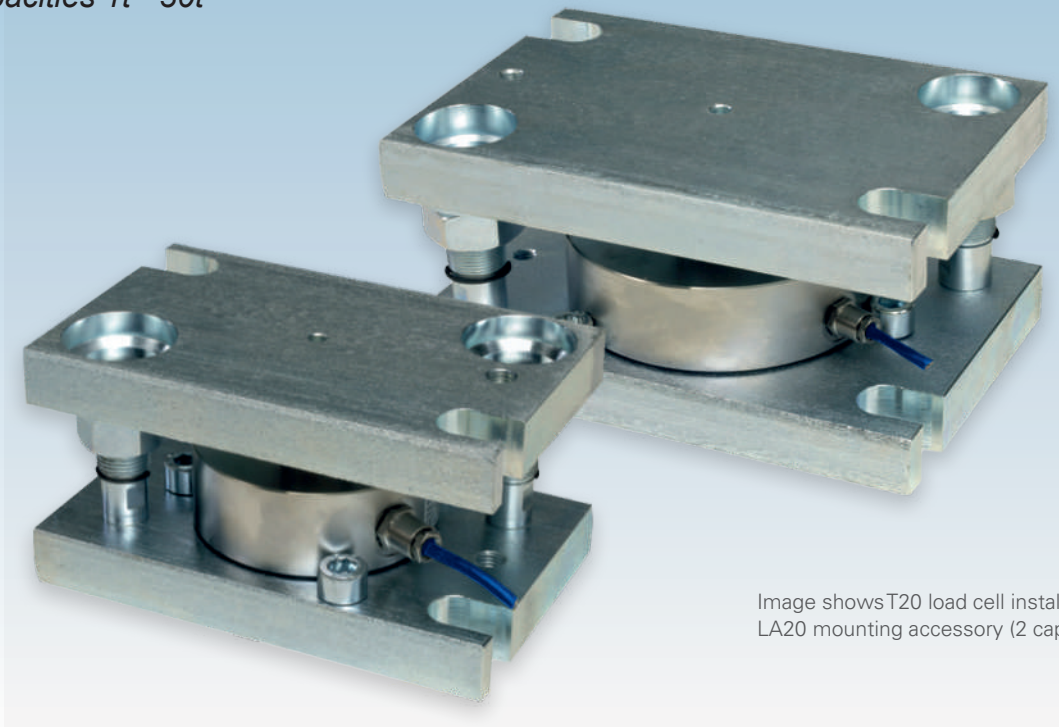


Image shows T20 load cell installed within LA20 mounting accessory (2 capacity ranges)

The T20 with LA20 weighing assembly is specifically designed as a reliable, simple, low profile and low cost solution for weighing of tanks, silos and vessels. It offers an excellent compromise between accuracy and cost (excellent price/quality ratio).

The stainless steel T20 load cell is mounted in a compact LA20 mounting accessory (either zinc plated alloy steel or stainless steel) that has integrated jacking nuts to allow simple, fast load cell installation and removal. The LA20 mounting accessory offers integrated lift-off protection and allowance for the angular misalignment that is often found in silos, tanks and other large metal structures. The mounting accessory incorporates a hardened alloy steel or stainless steel load bearing disc, for optimum application of force to the T20 load cell.

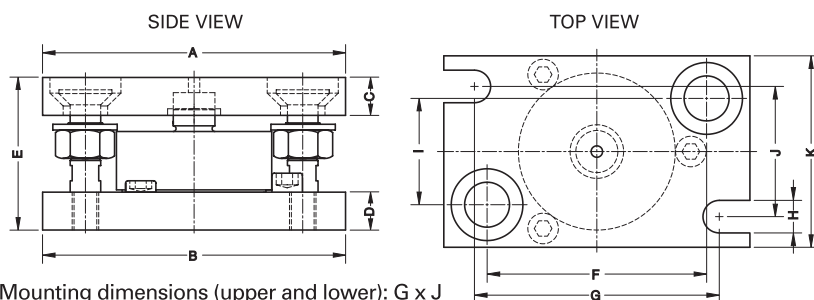
The T20 load cell is suitable for use in a fully live system with working load cells under all legs/support points of the vessel, or – where cost is more important than accuracy – use in a live/dummy system with either 1 live and 2 dummy (total 3) load cells or 2 live and 2 dummy (total 4) load cells.

- Stainless steel compression load cell
- Fully welded and hermetically sealed construction, protected to IP68 and IP69K
- Allows for thermal expansion and contraction of vessel
- Integrated threaded nuts to lift the top plate without hydraulic jacks or lifting tools
- Combined Error < $\pm 0.025\%$
- Low profile, compact mounting accessory
- Integrated lift-off protection
- Allowance for angular mis-alignment
- 5 Year Warranty
- Durable polyurethane cable with high resistance to chemicals, impacts, UV radiation and rodents.
- Option of High Temperature, or Parylene coated, or  and  certified versions



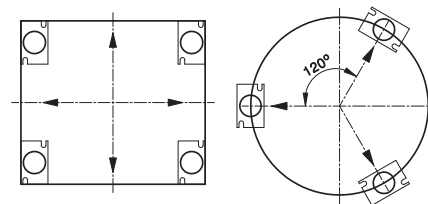
T20 with LA20

technical specification...



Mounting dimensions (upper and lower): G x J

Mounting Proposal Examples



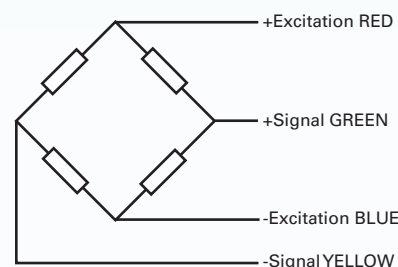
T20 Load Cell

	Load cell specification	Units
Load Cell Capacities (E_{max})	1, 2.5, 5, 10, 20, 30	tonnes (t)
Rated Output (S_n)	2.0	mV/V $\pm 0.1\%$
Combined Error	$< \pm 0.025$	% S_n
Non-Repeatability	$< \pm 0.015$	% S_n
Minimum load cell verification interval (V_{min}) = E_{max}/Y	$E_{max} / 10000$	kg
Creep (30 minutes)	$< \pm 0.024$	% S_n
Temperature Effect on Zero Balance	$< \pm 0.002$	% $S_n / ^\circ C$
Temperature Effect on Span	$< \pm 0.0018$	% $S_n / ^\circ C$
Compensated Temperature Range	-10 to +40	$^\circ C$
Operating Temperature Range	-50 to +70	$^\circ C$
Safe Load Limit (E_{lim})	200	% S_n
Zero Balance	± 2.0	% S_n
Input Resistance	800 ± 30	Ω
Output Resistance	700 ± 5	Ω
Insulation Resistance	> 5000	M Ω
Ingress Protection Rating	IP68 and IP69K	
Cable Length	10	m
Maximum Deflection at E_{max}	< 0.6	mm
Nominal Shipping Weight	1t-10t	7.0 kg
(Load Cell with Accessory)	20t-30t	15.8 kg

Dimensions

Capacity (t)	1, 2.5, 5, 10	20, 30
A, B	160	218
C, D	20	25
E	80	100
F	115	168
G	128	180
H	17	21
I	55.5	100
J	68	100
K	100	150

All dimensions are in mm



Weighing Assembly Data

Mounting Accessory	Load Cell Capacity (t)	Maximum Upwards (Lift-off) Load (kg)	Maximum Side or End Load (kg)	Maximum Side Offset (mm)
LA20-10T-ZINC LA20-10T-SS	1, 2.5, 5, 10	7339	2243	± 3
LA20-30T-ZINC LA20-30T-SS	20, 30	12644	3262	± 3

The above data is applicable with or without the load cell installed.

Model T20 ATEX and IECEx Certification

Certification Type	EX Marking	Zone Application
ATEX	II 1 GD Ex ia IIC T4...T6 Ga Ex ia IIIC T85°C Da Ex ta IIIC T85°C Da	Gas zones 0, 1, 2 with safety barriers (Ex ia) Dust zones 20, 21, 22 without safety barriers (Ex ta)
	Ex ia IIC T4...T6 Ga Ex ia IIIC T200 85°C Da	Gas zones 0, 1, 2 with safety barriers (Ex ia) Dust zones 20, 21, 22 with safety barriers (Ex ia)
IECEx	Ex ta IIIC T200 85°C Da ($-20^\circ C \leq T_{amb} \leq 40^\circ C$) Ex ta IIIC T200 105°C Da ($-20^\circ C \leq T_{amb} \leq 60^\circ C$)	Dust zones 20, 21, 22 without safety barriers (Ex ta)
	Ex ec IIC T6 Gc	Gas zone 2 without safety barriers (Ex ec)
	Ex tc IIIC T85°C Dc ($-20^\circ C \leq T_{amb} \leq 60^\circ C$)	Dust zone 22 without safety barriers (Ex tc)

DISTRIBUTED BY:

Electrical Connections

Via 4 core, 5.7mm diameter, screened Polyurethane cable
Screen not connected electrically to load cell.

Construction

Load cell T20
Stainless steel

Mounting accessory LA20

- Zinc plated version LA20-***-ZINC
Upper plate, lower plate, hardware: zinc plated alloy steel
- Stainless steel version LA20-***-SS
Upper plate, lower plate, hardware: stainless steel

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Our policy is one of continuous product enhancement. We therefore reserve the right to incorporate technical modifications without prior notification.



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